Lecture 8: Full MCQ Collection

1. What does the Wiedemann-Franz Law relate?

a. The elastic modulus and yield strength of materials.

b. The strength and density of materials.

c. The electrical and thermal conductivities of materials.

d. The toughness and thermal conductivity of materials.

Answer: c. The electrical and thermal conductivities of materials.

2. Which of the following is NOT a strengthening mechanism in metals?

a. Work hardening

b. Precipitation hardening

c. Surface coating

d. Solid solution hardening

Answer: c. Surface coating

3. What type of bonding leads to the highest modulus in materials?

a. Van der Waals bonding

b. Metallic bonding

c. Ionic or covalent bonding

d. Hydrogen bonding

Answer: c. Ionic or covalent bonding

4. Which structural design is used in foams to reduce weight while maintaining strength?

a. Crystalline grains

b. Cell structures

c. Lamellar layers

d. Glassy phases

Answer: b. Cell structures

5. What does quenching achieve in heat-treated steels?

a. Reduces modulus

b. Promotes grain growth

c. Forms martensite for increased strength

d. Removes dislocations

Answer: c. Forms martensite for increased strength

6. What does solid solution hardening involve?

a. Increasing grain size to reduce boundaries

b. Adding larger or smaller solute atoms to distort the lattice

c. Precipitating fine particles within the grain structure

d. Heating the metal above its melting point

Answer: b. Adding larger or smaller solute atoms to distort the lattice

7. Which material class generally has the highest modulus?

a. Foams

b. Polymers

c. Ceramics

d. Elastomers

Answer: c. Ceramics

8. Why do foams have low stiffness compared to solid materials?

a. Because they are made of brittle glass

b. Because of weak atomic bonding

c. Because they contain a high volume of air

d. Because they are highly crystalline

Answer: c. Because they contain a high volume of air

9. What is the main reason ceramics are strong but brittle?

a. They have metallic bonding

b. They have few dislocations and strong ionic/covalent bonds

c. They are composed of flexible polymers

d. Their atomic structure contains layers that slip easily

Answer: b. They have few dislocations and strong ionic/covalent bonds

10. Which material property is most affected by atomic bonding type?

a. Thermal diffusivity

b. Density

c. Modulus (stiffness)

d. Grain size

Answer: c. Modulus (stiffness)

11. What feature of composites improves their strength-to-weight ratio?

a. Hollow internal pores

b. Dense crystalline regions

c. Fibre reinforcement in a matrix

d. High metal content

Answer: c. Fibre reinforcement in a matrix

12. In precipitation hardening, what is the role of aging?

a. To increase grain size for flexibility

b. To dissolve precipitates into the matrix

c. To form tiny particles that block dislocations

d. To anneal out residual stress

Answer: c. To form tiny particles that block dislocations

13. What happens if you cool metal too slowly during heat treatment?

a. You form brittle martensite

b. You form softer phases like pearlite

c. Dislocations are removed

d. The metal turns into ceramic

Answer: b. You form softer phases like pearlite